EDMS 657 HW2

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2025-09-24

Part A

1) (1 point)

What is the maximum number of principal components that can be extracted from these data? Explain.

The maximum number of principal components would be 22, since there are 22 variables. It could be that all variables are important contributors in their own right, however unlikely this may be.

2) (4 points)

Using the software of your choice, run a PCA based on the correlation matrix extracting all possible principal components (also request a scree plot), but without any rotation. Also get a component score coefficient matrix. Append these results to the end of your homework.

3) (1 point)

How many units of variance does the first component explain in the original 22 (standardized) variables? Explain the source of your answer.

```r
pc_analysis$Vaccounted[1,1]
```

```
## [1] 5.596377
```

The first component explains 5.6 units of variance (rounded from 5.596). This is given by the first SS loading in the principal() output. Since the original variables were standardized, we can take their total variance to be the number of variables (each with one unit of variance). This is convenient because it means we do not have to convert the output of the function.

4) (1 point)

According to the component score coefficients, what is the "recipe" for building the first component? Write out the equation for Z1 in terms of the X variables (which are assumed to be standardized). Because this equation would have 22 predictors, write out the equation with only the first 5 variables (pubspeak, law, busmgmt, sales, merch).

```
pc_analysis$weights[,1] %>% head(5)
```

```
## [1] 0.1292509 0.1098682 0.1448036 0.1390747 0.1474039
```

$PC_1 = 0.1292 pubspeak + 0.1098 law + 0.1448 busmgmt + 0.1391 sales + 0.1474 merch + ...$

5) (1 point)

According to the loadings in the component matrix, what is the "recipe" for rebuilding pubspeak based on scores from the 22 principal components extracted? Write out the equation for pubspeak with only the first 5 components (Z1, Z2, . . . , Z5), which are assumed to be standardized.

```
pc_analysis$loadings[1,] %>% head(5)
```

```
##         PC1         PC2         PC3         PC4         PC5
##  0.72333666  0.08700573 -0.38915460  0.06515927  0.25714472
```

$pubspeak = 0.72 Z_1 + 0.09 Z_2 - 0.39 Z_3 + 0.07 Z_4 + 0.26 Z_5 + ...$

6) (1 point)

Derive the first eigenvalue using the appropriate loadings, showing your work.

```
pc_analysis$loadings[,1]^2 %>% sum()
```

```
## [1] 5.596377
```

```
pc_analysis$values[1]
```

```
## [1] 5.596377
```

The eigenvalues are obtained by summing the squares of the loadings on the corresponding component. Thus, the above code takes the first column of the loadings matrix, squares them, then adds all the resulting values together to come up with the first eigenvalue. The second line of code extracts the value directly from the principal() output, for verification.

7)

a. (1 point)

How many components would need to be extracted in order to extract 80% of the variance in the original standardized X variables?

```r
which(pc_analysis$Vaccounted[5,] > 0.8)[1] # first component where % variance >80
```

```
## PC8
##   8
```

8 components need to be extracted to explain 80% of the variance in the original data, which is given by the "Cumulative Var" or "Cumulative Proportion" rows in the principal() output. These values are the same because all x variables were standardized.

b. (1 point)

How many components would be extracted using Kaiser's rule?

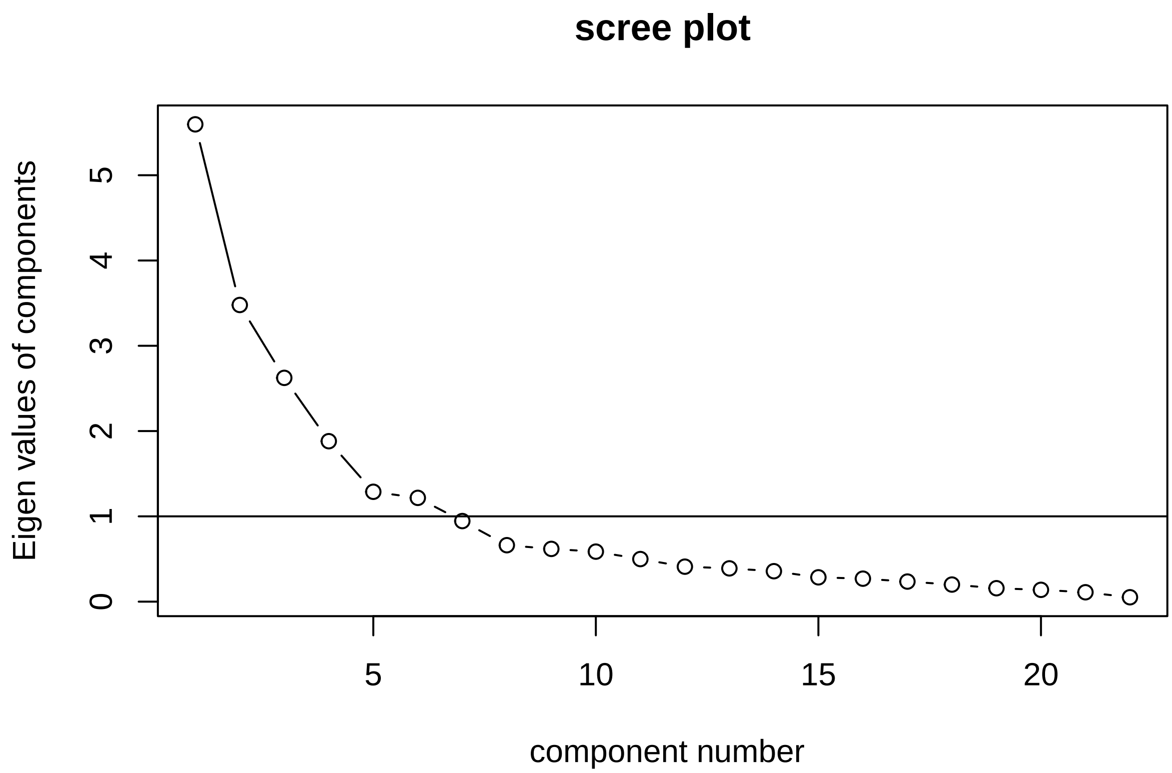
```r
pc_analysis$values
```

```
##  [1] 5.59637727 3.47853135 2.62393808 1.88134833 1.28794966 1.21677289
##  [7] 0.94491507 0.66186961 0.61826691 0.58644127 0.49916135 0.41020471
## [13] 0.39015558 0.35692400 0.28509856 0.26947987 0.23531220 0.20017131
## [19] 0.15734287 0.13834357 0.10983372 0.05156184
```

Kaiser's rule keeps all components with an eigenvalue greater than 1, so we would keep the first 6 components as the 7th has an eigenvalue of 0.94. This is also given by the principal() output, looking at the SS loadings row. We can also see this in the following scree plot.

c. (1 point)

Based on the scree plot, how many components do you think should be extracted? Paste scree plot here and explain.



Based on the scree plot, I would probably keep 5 principal components as that's where the "elbow" point appears to be. In other words, each successive component contributes a relatively small amount of the overall variance (they are not contributing that much, whereas the first 5 are contributing a fair amount).

d. (3 points)

Perform Velicer's Minimum Average Partial (MAP) procedure to see how many components should be extracted; paste supporting output into your response.

```
velicer_map <- VSS(cormat,rotate='none',plot=F,n.obs=231)
velicer_map
```

```
##
## Very Simple Structure
## Call: vss(x = x, n = n, rotate = rotate, diagonal = diagonal, fm = fm,
##     n.obs = n.obs, plot = plot, title = title, use = use, cor = cor)
## VSS complexity 1 achieves a maximimum of 0.61  with  4  factors
## VSS complexity 2 achieves a maximimum of 0.76  with  4  factors
##
## The Velicer MAP achieves a minimum of 0.04  with  6  factors
## BIC achieves a minimum of  -268.95  with  7  factors
## Sample Size adjusted BIC achieves a minimum of  28.11  with  8  factors
##
## Statistics by number of factors
##   vss1 vss2   map dof chisq     prob sqresid  fit RMSEA     BIC SABIC complex
```

```
## 1 0.51 0.00 0.066 209   2168 2.6e-322    29.2 0.51 0.201 1030.39  1693     1.0
## 2 0.57 0.71 0.047 188   1475 2.4e-198    17.3 0.71 0.172  451.93  1048     1.5
## 3 0.61 0.75 0.042 168   1088 1.8e-134    10.5 0.82 0.154  173.75   706     1.8
## 4 0.61 0.76 0.039 149    810  1.7e-91     7.1 0.88 0.139   -0.75   471     2.0
## 5 0.61 0.74 0.037 131    530  1.9e-49     5.5 0.91 0.115 -182.66   233     2.4
## 6 0.61 0.73 0.036 114    383  8.0e-31     4.2 0.93 0.101 -237.27   124     2.6
## 7 0.61 0.73 0.038  98    264  3.2e-17     3.3 0.95 0.086 -268.95    42     2.8
## 8 0.61 0.74 0.046  83    217  6.7e-14     2.7 0.95 0.083 -234.95    28     2.9
##   eChisq  SRMR eCRMS eBIC
## 1   3466 0.180 0.189 2329
## 2   1642 0.124 0.137  619
## 3    760 0.084 0.099 -154
## 4    380 0.060 0.074 -431
## 5    216 0.045 0.060 -497
## 6    101 0.031 0.044 -519
## 7     47 0.021 0.032 -486
## 8     35 0.018 0.030 -416
```

The beginning of the output suggests that 6 factors (components) should be utilized, as the Velicer MAP procedure achieved a minimum (partial correlation) of 0.04 with that many factors included.

8) (2 points)

Assume the researcher wants the first six components. Based on the loadings on these six components, compute the communality and uniqueness for pubspeak.

```
commun <- sum(pc_analysis$loadings[1,1:6]^2) %>% round(3)
uniqueness <- (1-commun) %>% round(3)
paste("Communality for pubspeak from first six components:", commun,
      "Uniqueness:", uniqueness)
```

```
## [1] "Communality for pubspeak from first six components: 0.794 Uniqueness: 0.206"
```

9) (2 points)

Based on the loadings for the first six components, manually compute the reproduced correlation between pubspeak and law, showing your work. Also compute the residual correlation not explained by these six components.

```
reproduced_cor <- (principal(cormat,nfactors=6,rotate='none')$loadings %>% factor.model())[1,2]
rho_pub_law <- cormat[1,2]
residual <- reproduced_cor - rho_pub_law
paste("The reproduced correlation between pubspeak and law is", reproduced_cor)
```

```
## [1] "The reproduced correlation between pubspeak and law is 0.723618229275963"
```

```
paste("Residual correlation not explained by the components is", residual)
```

```
## [1] "Residual correlation not explained by the components is -0.0463817707240373"
```

Part B (12 points)

1) (3 points)

Assuming the researcher wants the first six components, extract those six components and rotate them using Varimax rotation. As part of your output be sure to get the initial solution, the rotated solution, and a rotated component score coefficient matrix. No scree plot is necessary. Edit your results down to no more than three pages (shrinking and positioning things as necessary) and insert/paste those pages here, not at the end.

```r
pc_analysis <- principal(cormat,nfactors=6,rotate="none")
pc_analysis2 <- principal(cormat,nfactors=6)
pc_analysis # initial solution
```

```
## Principal Components Analysis
## Call: principal(r = cormat, nfactors = 6, rotate = "none")
## Standardized loadings (pattern matrix) based upon correlation matrix
##     PC1   PC2   PC3   PC4   PC5   PC6   h2   u2 com
## 1  0.72  0.09 -0.39  0.07  0.26  0.20 0.79 0.21 2.1
## 2  0.61 -0.10 -0.37  0.09  0.45  0.11 0.75 0.25 2.8
## 3  0.81 -0.42 -0.08 -0.16 -0.13  0.09 0.89 0.11 1.7
## 4  0.78 -0.27 -0.14 -0.12 -0.12  0.11 0.74 0.26 1.5
## 5  0.82 -0.36 -0.12 -0.16 -0.18  0.15 0.90 0.10 1.7
## 6  0.68 -0.44  0.04 -0.25 -0.18 -0.22 0.79 0.21 2.5
## 7  0.40 -0.22  0.32  0.33  0.02 -0.31 0.52 0.48 4.5
## 8  0.69 -0.43  0.14 -0.13 -0.20  0.05 0.74 0.26 2.1
## 9  0.07 -0.13  0.58 -0.41  0.29  0.07 0.62 0.38 2.5
## 10 0.14  0.29  0.75 -0.28  0.24 -0.07 0.81 0.19 2.0
## 11 0.36  0.00  0.76 -0.18  0.01  0.05 0.74 0.26 1.6
## 12 0.36  0.56  0.41  0.33 -0.34  0.02 0.84 0.16 4.2
## 13 0.27  0.23  0.34  0.65 -0.41  0.17 0.86 0.14 3.2
## 14 0.26 -0.05  0.28  0.42  0.33  0.54 0.73 0.27 3.8
## 15 0.34 -0.22  0.12  0.61  0.21 -0.09 0.61 0.39 2.3
## 16 0.35  0.33  0.25 -0.02  0.47 -0.15 0.54 0.46 3.6
## 17 0.43  0.42 -0.31  0.17  0.21 -0.41 0.71 0.29 4.6
## 18 0.48  0.37 -0.01  0.07 -0.04 -0.44 0.57 0.43 3.0
## 19 0.58  0.40 -0.07 -0.08 -0.07 -0.31 0.61 0.39 2.5
## 20 0.26  0.74 -0.08 -0.39 -0.08  0.14 0.79 0.21 2.0
## 21 0.23  0.78  0.01 -0.22 -0.17  0.28 0.81 0.19 1.8
## 22 0.35  0.65 -0.38  0.01  0.03  0.23 0.74 0.26 2.5
##
##                        PC1  PC2  PC3  PC4  PC5  PC6
## SS loadings           5.60 3.48 2.62 1.88 1.29 1.22
## Proportion Var        0.25 0.16 0.12 0.09 0.06 0.06
## Cumulative Var        0.25 0.41 0.53 0.62 0.68 0.73
## Proportion Explained  0.35 0.22 0.16 0.12 0.08 0.08
## Cumulative Proportion 0.35 0.56 0.73 0.84 0.92 1.00
##
## Mean item complexity =  2.6
## Test of the hypothesis that 6 components are sufficient.
##
## The root mean square of the residuals (RMSR) is  0.05
##
```

```
## Fit based upon off diagonal values = 0.97
```

```
pc_analysis2 # rotated solution
```

```
## Principal Components Analysis
## Call: principal(r = cormat, nfactors = 6)
## Standardized loadings (pattern matrix) based upon correlation matrix
##      RC1   RC2   RC6   RC3   RC4   RC5   h2   u2 com
## 1   0.49  0.26  0.34 -0.16 -0.06  0.58 0.79 0.21 3.3
## 2   0.43  0.02  0.31 -0.12 -0.22  0.64 0.75 0.25 2.7
## 3   0.93 -0.03  0.07  0.03  0.01  0.15 0.89 0.11 1.1
## 4   0.82  0.07  0.13 -0.03  0.03  0.18 0.74 0.26 1.2
## 5   0.93  0.05  0.06 -0.01  0.04  0.15 0.90 0.10 1.1
## 6   0.83 -0.19  0.19  0.13 -0.06 -0.13 0.79 0.21 1.3
## 7   0.27 -0.46  0.30  0.20  0.32  0.08 0.52 0.48 3.9
## 8   0.82 -0.12 -0.01  0.16  0.12  0.03 0.74 0.26 1.2
## 9   0.10 -0.04 -0.16  0.75 -0.17  0.01 0.62 0.38 1.2
## 10 -0.09  0.11  0.14  0.87  0.11 -0.03 0.81 0.19 1.1
## 11  0.26 -0.01 -0.02  0.76  0.29 -0.02 0.74 0.26 1.6
## 12  0.00  0.28  0.30  0.20  0.79 -0.02 0.84 0.16 1.7
## 13  0.04 -0.01  0.04 -0.04  0.91  0.12 0.86 0.14 1.0
## 14  0.05 -0.05 -0.22  0.21  0.34  0.72 0.73 0.27 1.9
## 15  0.13 -0.49  0.22 -0.03  0.33  0.44 0.61 0.39 3.4
## 16 -0.04  0.09  0.47  0.45 -0.01  0.32 0.54 0.46 2.9
## 17  0.03  0.13  0.79 -0.16  0.01  0.19 0.71 0.29 1.3
## 18  0.16  0.11  0.70  0.05  0.19 -0.06 0.57 0.43 1.4
## 19  0.29  0.29  0.65  0.07  0.13 -0.05 0.61 0.39 2.0
## 20  0.00  0.82  0.30  0.16  0.02 -0.05 0.79 0.21 1.3
## 21 -0.05  0.84  0.19  0.13  0.24  0.01 0.81 0.19 1.3
## 22  0.02  0.69  0.35 -0.22  0.10  0.29 0.74 0.26 2.2
##
##                        RC1  RC2  RC6  RC3  RC4  RC5
## SS loadings           4.49 2.64 2.59 2.43 2.12 1.82
## Proportion Var        0.20 0.12 0.12 0.11 0.10 0.08
## Cumulative Var        0.20 0.32 0.44 0.55 0.65 0.73
## Proportion Explained  0.28 0.16 0.16 0.15 0.13 0.11
## Cumulative Proportion 0.28 0.44 0.60 0.76 0.89 1.00
##
## Mean item complexity =  1.8
## Test of the hypothesis that 6 components are sufficient.
##
## The root mean square of the residuals (RMSR) is  0.05
##
## Fit based upon off diagonal values = 0.97
```

```
pc_analysis2$weights # rotated component score coefficient matrix
```

```
##               RC1          RC2          RC6          RC3          RC4
##  [1,]  0.053869313  0.093625742  0.019210374 -0.055811322 -0.087425140
##  [2,]  0.010127853 -0.015164985  0.071710538 -0.008801243 -0.183842980
##  [3,]  0.230037942  0.030628689 -0.072348018 -0.018017426 -0.003827785
##  [4,]  0.200589952  0.061014825 -0.055835065 -0.040561762  0.006470824
##  [5,]  0.239683603  0.072808838 -0.098890479 -0.039169882  0.018320914
```

```
##  [6,]  0.208373523 -0.077248915  0.081201689  0.028625102 -0.042609182
##  [7,]  0.002396860 -0.256210028  0.192164644  0.051967454  0.116454079
##  [8,]  0.215312623 -0.003770429 -0.082832334  0.027012530  0.060395517
##  [9,]  0.006358040  0.011989686 -0.069178477  0.344021188 -0.156968196
## [10,] -0.064586668  0.008410952  0.064868854  0.373044238 -0.052081732
## [11,]  0.051373700  0.009999288 -0.060628282  0.291501171  0.082123426
## [12,] -0.007894086  0.064847301  0.032421645 -0.001179677  0.375471801
## [13,]  0.007677895 -0.006463538 -0.097112049 -0.116235915  0.488992746
## [14,] -0.057338804  0.034016751 -0.240350706  0.086612205  0.129088086
## [15,] -0.064019249 -0.253088446  0.119827499 -0.027221752  0.127795166
## [16,] -0.113527984 -0.058015428  0.217858263  0.224000970 -0.134298925
## [17,] -0.084030678 -0.098569176  0.394330842 -0.055498831 -0.079645312
## [18,] -0.014449803 -0.084496844  0.341423690 -0.002342223  0.030415078
## [19,]  0.034368320  0.014099704  0.267527982  0.005304693  0.003189214
## [20,]  0.021468320  0.315933586  0.002248260  0.058112639 -0.034668690
## [21,]  0.016912231  0.342777842 -0.088977846  0.025649693  0.096038559
## [22,] -0.014075876  0.252446340  0.006771157 -0.096714866  0.017242009
##               RC5
##  [1,]  0.292473709
##  [2,]  0.357208013
##  [3,] -0.016941737
##  [4,]  0.005448923
##  [5,] -0.019026588
##  [6,] -0.205014040
##  [7,] -0.038268648
##  [8,] -0.083103037
##  [9,]  0.069008300
## [10,]  0.013219127
## [11,] -0.029623904
## [12,] -0.099347883
## [13,] -0.001371522
## [14,]  0.493654213
## [15,]  0.222812107
## [16,]  0.191743758
## [17,]  0.021611030
## [18,] -0.156132991
## [19,] -0.146791037
## [20,] -0.052067192
## [21,] -0.007787552
## [22,]  0.147437820
```

2) (2 points)

Focusing on the loadings of the original variables on the six rotated principal components, explain how this pattern of loadings represents a clearer "simple structure" than the unrotated solution.

Rotation represents a clearer simple structure because there is less mixing of signs and magnitudes. For example, The second unrotated principal component is strongly (with 0.4 in absolute value being the arbitrary threshold) *positively* loaded on nature, social service, teaching, music, art, and writing, but strongly *negatively* loaded on business management, office practice, and technical supervision. It is difficult to come up with a phrase that captures this "skillset". However, the rotated second component is strongly positively loaded on music, art, writing, and only strongly negatively loaded on military activities and recreational leadership. Firstly, there are less "strong" loadings, which gives the rotated component more specificity than the principal component (which is more of a "general" summary of all the variables). Secondly, we are able to pick out specific patterns in skills more easily (see the last problem which attempts to give "names" to the rotated components).

This can also be seen by comparing the first principal component to the first rotated component. The first PC is strongly positively loaded on *most* traits, and is thus not very useful in determining a specific factor. However, the first RC is strongly positively loaded on variables 1 through 6 and 8 only. Although the rotated component does not capture as much *overall* variance as the principal component, it offers a simpler structure and thus facilitates interpretation.

3) (1 point)

According to the rotated component score coefficient matrix, what is the "recipe" for building the first rotated component? Write out the equation for Z_1^* in terms of the X variables (which are assumed to be standardized). Because this equation would have 22 predictors, write out the equation with only the first 5 variables (pubspeak, law, busmgmt, sales, merch).

```
pc_analysis2$loadings[1:5,1]
```

```
## [1] 0.4948655 0.4251593 0.9266385 0.8244402 0.9318801
```

$RC_1 = 0.495 pubspeak + 0.425 law + 0.927 busmgmt + 0.824 sales + 0.932 merch + ...$

4) (1 point)

Derive the "rotated eigenvalue" of the first rotated component using the appropriate rotated loadings, showing your work.

```
pc_analysis2$loadings[,1]^2 %>% sum()
```

```
## [1] 4.489279
```

```
pc_analysis2$Vaccounted[1,1]
```

```
## [1] 4.489279
```

This was done to a similar way as before - summing the squares of the first column of loadings. The second line is once again included to verify the calculated. The actual eigenvalues are still the same as before, but the sum of squared loadings can be accessed through the Vaccounted element in the principal() output list.

5) (1 point)

Show that the six rotated principal components account for the same total amount of variability in the original 22 (standardized) variables as did the first six un-rotated principal components.

```r
pc_analysis$Vaccounted[1,] %>% sum()
```

```
## [1] 16.08492
```

```r
pc_analysis2$Vaccounted[1,] %>% sum()
```

```
## [1] 16.08492
```

The later *rotated* components contribute a higher proportion of the variance explained than the later *principal* components, but the same amount of variance is explained in total. In other words, the variance of the loadings is smaller for the rotated components than the principal components.

6) (2 points)

Using the original loadings and then the rotated loadings, show that the six rotated components explain the same amount of variability in pubspeak as did the six un-rotated components (i.e., computationally verify the equality of its communality before and after rotation of the six components).

```r
sum(pc_analysis$loadings[1,]^2)
```

```
## [1] 0.7939732
```

```r
sum(pc_analysis2$loadings[1,]^2)
```

```
## [1] 0.7939732
```

The above code sums the squares from the first row of the loadings from each analysis (as is the formula for communality), and finds them to be equal.

7) (2 points)

Based on the larger loadings (e.g., absolute value >.4), how would you interpret the first six rotated components? That is, what names might you give to them? Explain. Do your best to make sure that each factor name sounds like something that a person could have less or more of (i.e., a continuum).

None of this is to make overgeneralizations about people or the skills they have.

- RC1 is positive on public speaking, law and politics, business management, sales, merchandising, office practice, technical supervision, negative on no others. I would name this **entrepreneurial**, or business-orientation. Business, sales, office practice, supervision are all related to business-oriented skills, of course, and public speaking seems to fall right into that. I imagine someone who would score highly on this component would be a manager, or even a CEO of a big company. They have to have the lower level skills like sales and merchandising, but also be good at public speaking to communicate with their workforce, and good at law and politics to ensure the best business practices (as law/politics play a huge role in business).

- RC2 is positive on music, art, and writing, negative on military activities, recreational leadership. This strikes me as **"artfulness"**, because each variable included are fine arts / literature. Military activities and recreational leadership are not generally associated with artfulness, and it seems common that more militaristic people tend to be less artsy and more strategic.
- RC6 is positive on medical service, social service, religious activities, teaching. **Helpfulness**, or servitude, makes sense for this one because all variables in some way relate to helping people, or using your skills to make a difference on the world. It can also be pretty indicative of social ability, since each variable involves a lot of dealing with other people.
- RC3 is positive on math, science, mechanical, medical service. **Technical skill**, or some sort of STEM ability. These subjects are fairly analytical, stereotypically "left brain", and are usually viewed as "technical" skills. That is, often when you say someone is being technical, they're probably talking about something in math rather than something in painting (although both are very technical in their own ways), at least generally.
- RC4 is positive on nature and agriculture. **Outdoors-y-ness**, or just "knowledge of the natural world" makes a lot of sense here. The variables aren't necessarily strongly related to social skills nor are they notoriously rigorous academic subjects.
- RC5 positive on public speaking, law and politics, adventure, recreational leadership. One might name this **charisma**, because public speaking and law/politics each require verbal communication skills, and generally speakers, politicians, and leaders with more charisma are more known. Adventure seems like a natural extension of this, adventurous people tend to be extroverted and thus charismatic. (Alternatively, the Teddy Roosevelt index)

Appended Work:

Part A

2

```r
pc_analysis$loadings
```

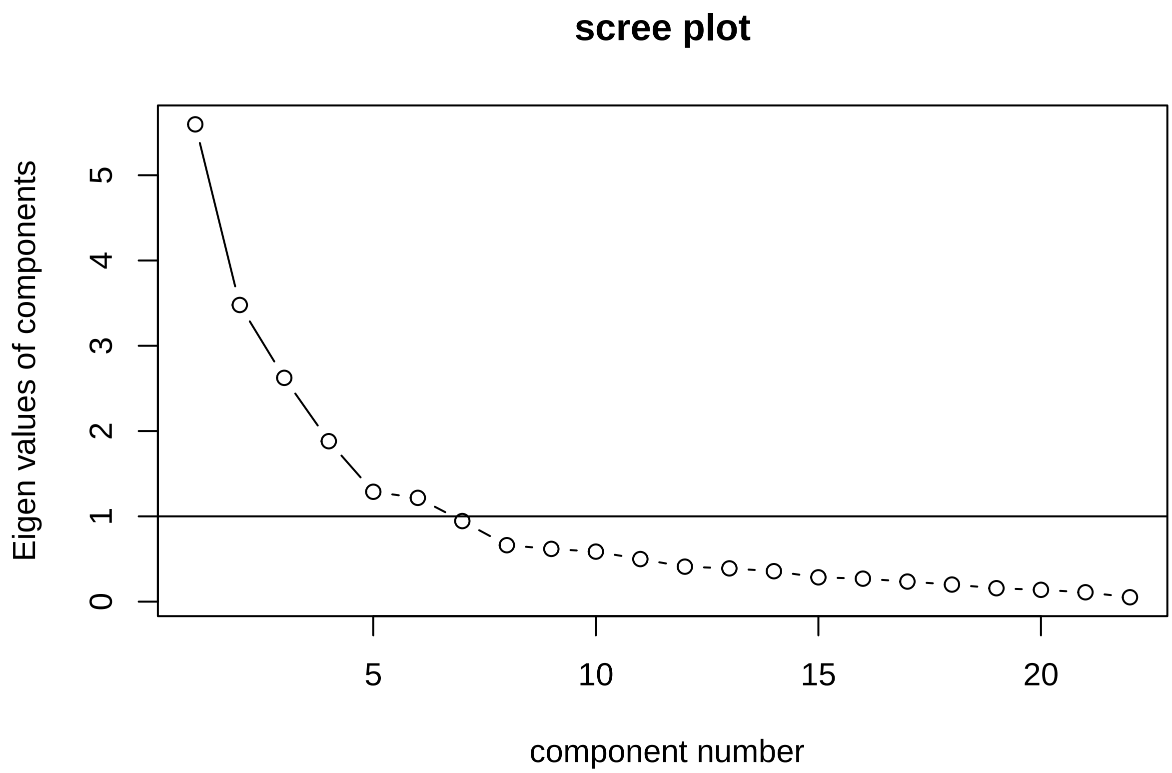
```
##
## Loadings:
##       PC1    PC2    PC3    PC4    PC5    PC6
##  [1,]  0.723        -0.389         0.257  0.203
##  [2,]  0.615 -0.102 -0.366         0.453  0.113
##  [3,]  0.810 -0.418        -0.159 -0.130
##  [4,]  0.778 -0.266 -0.143 -0.120 -0.118  0.110
##  [5,]  0.825 -0.357 -0.117 -0.164 -0.176  0.146
##  [6,]  0.675 -0.435        -0.252 -0.176 -0.223
##  [7,]  0.403 -0.218  0.322  0.326        -0.312
##  [8,]  0.688 -0.426  0.140 -0.133 -0.202
##  [9,]        -0.131  0.585 -0.411  0.289
## [10,]  0.138  0.287  0.749 -0.281  0.244
## [11,]  0.358         0.758 -0.182
## [12,]  0.362  0.560  0.409  0.331 -0.342
## [13,]  0.268  0.227  0.343  0.648 -0.406  0.174
## [14,]  0.261         0.281  0.423  0.332  0.543
## [15,]  0.341 -0.223  0.121  0.615  0.211
## [16,]  0.351  0.329  0.252         0.467 -0.148
## [17,]  0.435  0.423 -0.312  0.174  0.208 -0.408
## [18,]  0.480  0.373                      -0.435
```

```
## [19,]  0.578  0.400                      -0.310
## [20,]  0.256  0.736        -0.395         0.136
## [21,]  0.235  0.776        -0.221 -0.167  0.282
## [22,]  0.350  0.650 -0.382                0.227
##
##                  PC1   PC2   PC3   PC4   PC5   PC6
## SS loadings    5.596 3.479 2.624 1.881 1.288 1.217
## Proportion Var 0.254 0.158 0.119 0.086 0.059 0.055
## Cumulative Var 0.254 0.412 0.532 0.617 0.676 0.731
```

```r
pc_analysis$r.scores
```

```
##               PC1           PC2           PC3           PC4           PC5
## PC1  1.000000e+00 -1.379452e-14  1.165734e-15 -7.415943e-16 -4.388850e-16
## PC2 -1.381534e-14  1.000000e+00 -7.771561e-16 -4.336809e-17 -1.085937e-15
## PC3  1.262879e-15 -7.216450e-16  1.000000e+00 -5.702903e-17 -7.233797e-16
## PC4 -7.302102e-16  4.553649e-17  3.144186e-18  1.000000e+00  4.504860e-17
## PC5 -4.516786e-16 -1.061651e-15 -7.424616e-16  1.067939e-17  1.000000e+00
## PC6  4.926615e-16 -3.747003e-16 -6.175616e-16  3.380542e-16  5.087077e-15
##               PC6
## PC1  4.163336e-16
## PC2 -3.747003e-16
## PC3 -5.689893e-16
## PC4  3.825065e-16
## PC5  5.195497e-15
## PC6  1.000000e+00
```

```r
VSS.scree(cormat)
```


scree plot
Eigen values of components
0
1
2
3
4
5
5
10
15
20
component number